

User Roles and Permissions Document

Rwanda NBSAP Monitoring Dashboard 2025–2030

Prepared for: Rwanda Environment Management Authority (REMA) **Document Reference:** NBSAP-URP-2026-007 **Version:** 1.0 **Date:** June 2026 **Classification:** Official — Government Use

Document Control

Field	Value
Document Title	User Roles and Permissions
Project	Rwanda NBSAP Monitoring Dashboard
Prepared by	NBSAP Dashboard Development Team
Approved by	REMA — Department of Biodiversity & Landscape Management
Status	Final

1. Overview

The NBSAP Monitoring Dashboard enforces a **6-tier role-based access control (RBAC)** model. Each user is assigned exactly one role that determines which pages they can access, which actions they can perform, and which data they can view. Access control is enforced at two levels:

- Client-side** — Protected routes check the user's role before rendering pages
- Server-side** — PostgreSQL Row-Level Security (RLS) policies enforce data access at the database level, preventing bypass via direct API calls

2. Role Definitions

2.1 REMA Administrator (`dashboard_management`)

Attribute	Value
Label	REMA Administrator

Typical Users	REMA national team, system administrators
Description	Full system access: verification queue, user management, audit log, all exports
User Count	2–5

Responsibilities:

- Manage all user accounts (create, suspend, deactivate, change roles)
- Review and approve/reject all submitted reports in the verification queue
- Access and audit the immutable system audit log
- Export raw data in all formats (CSV, PDF, JSON)
- Configure system settings and tool weights
- Monitor system health and data pipeline status
- Prepare data exports for CBD National Reporting

2.2 Lead Government Ministry Reporter (`lead_government_ministry_reporting`)

Attribute	Value
Label	Lead Government Ministry Reporter
Typical Users	Ministry focal points, sector ministry officers
Description	Submit reports via T01–T07, verify submissions, access analytics
User Count	10–20

Responsibilities:

- Submit biodiversity data via all 7 reporting tools (T01–T07)
- Review and approve/reject reports in the verification queue
- Access analytics and reporting dashboards
- View the risk register and compliance tracking
- Export data for sector-level analysis

2.3 District Reporting Officer (`local_reporting`)

Attribute	Value
Label	Local Reporter
Typical Users	District environmental officers (30 districts)
Description	Enter and validate district-level biodiversity monitoring data
User Count	30–60

Responsibilities:

- Submit reports for their assigned district
 - View their own submission history and status
 - Access compliance tracking for their district
 - View national dashboards and targets (read-only)
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2.4 Policy Monitor (`policy_monitoring`)

Attribute	Value
Label	Policy Monitor
Typical Users	National policymakers, Cabinet-level advisors, MoE oversight
Description	Read-only access to strategic dashboards and national progress summaries
User Count	5–15

Responsibilities:

- View executive dashboards and national progress
 - Monitor the risk register
 - Access compliance tracking summaries
 - View analytics and trend data
 - No data entry or modification capability
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2.5 Development Partner (`programme_alignment`)

Attribute	Value
Label	Development Partner
Typical Users	NGOs, research institutions, private sector partners, donors
Description	Analytical viewing access for programme alignment and donor reporting
User Count	10–30

Responsibilities:

- View analytical dashboards for programme alignment
 - Access the risk register and compliance data
 - View indicator and target progress
 - No data entry, export, or modification capability
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2.6 Public Viewer (`public_viewer`)

Attribute	Value
Label	Public Viewer
Typical Users	General public, media, academic researchers, civil society
Description	External read-only access to view dashboards, reports, indicators, and maps
User Count	Unlimited

Responsibilities:

- View public-facing dashboards
- View national targets and indicator progress
- View the interactive map
- No access to verification queue, audit log, user management, or raw exports

3. Permissions Matrix

3.1 Feature-Level Permissions

Permission	REMA Admin	Ministry Reporter	District Officer	Policy Monitor	Dev Partner	Public Viewer
Submit Reports (T01–T07)	Yes	Yes	Yes	No	No	No
Approve/Reject Reports	Yes	Yes	No	No	No	No
View Verification Queue	Yes	Yes	No	No	No	No
View Audit Log	Yes	No	No	No	No	No
Manage Users	Yes	No	No	No	No	No
Export Raw Data	Yes	Yes	No	No	No	No
View Risk Register	Yes	Yes	No	Yes	Yes	No
View Compliance	Yes	Yes	Yes	Yes	Yes	No
View Analytics	Yes	Yes	No	Yes	Yes	No

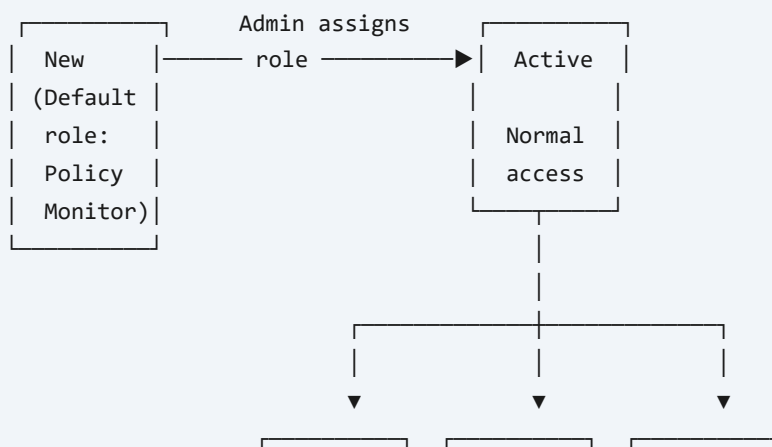
3.2 Page-Level Access

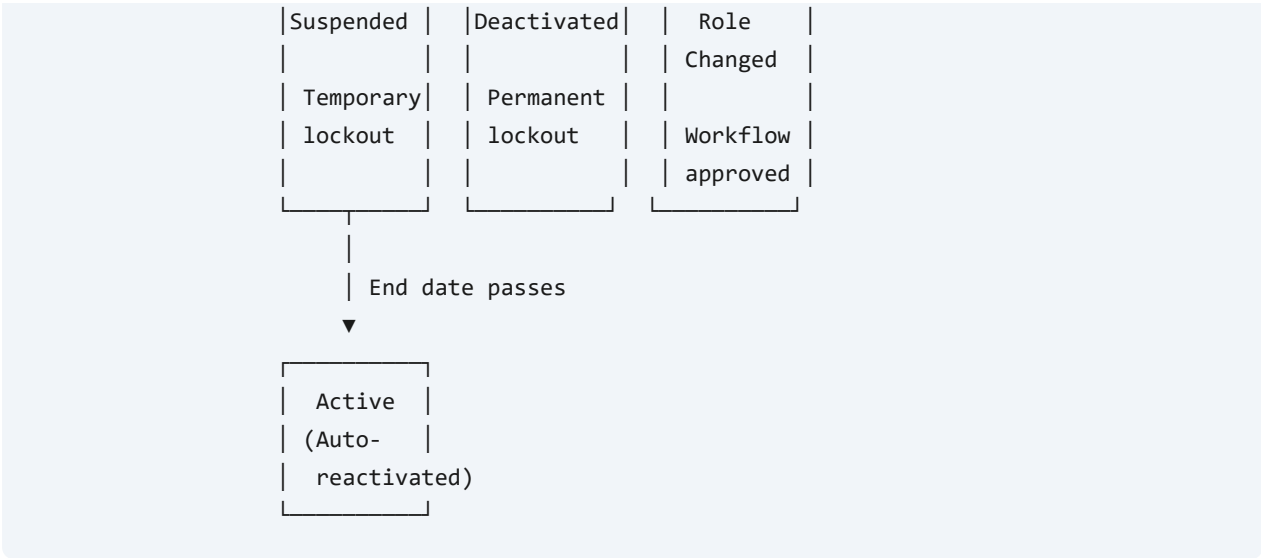
Page	REMA Admin	Ministry Reporter	District Officer	Policy Monitor	Dev Partner	Public Viewer
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Dashboard	Full	Full	Full	Full	Full	View
National Targets	Full	View	View	View	View	View
Indicators	Full	View	View	View	View	View
Interactive Map	Full	Full	Full	Full	Full	View
Reporting Toolkit	Full	Submit	Submit	No	No	No
Verification Queue	Full	Review	No	No	No	No
Reports & Analytics	Full	View	No	View	View	No
RBIS Dashboard	Full	View	View	View	View	View
Compliance	Full	View	View	View	View	No
Risk Register	Full	View	No	View	View	No
Stakeholders	Full	No	No	No	No	No
Data Pipeline	Full	No	No	No	No	No
User Management	Full	No	No	No	No	No
Role Requests	Full	No	No	No	No	No
Settings	Own	Own	Own	Own	Own	Own
Adaptive Management	Full	View	No	View	No	No

4. Account Lifecycle

4.1 Account States





4.2 Account Status Definitions

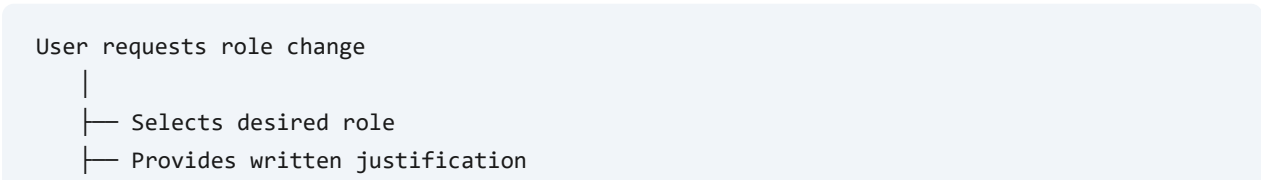
Status	Can Login	Description
Active	Yes	Normal operational status
Suspended	No	Temporary lockout with reason; auto-reactivates if end date set
Suspended (Expired)	Yes	Suspension end date has passed; auto-reactivation pending
Deactivated	No	Permanent lockout; requires admin to reactivate

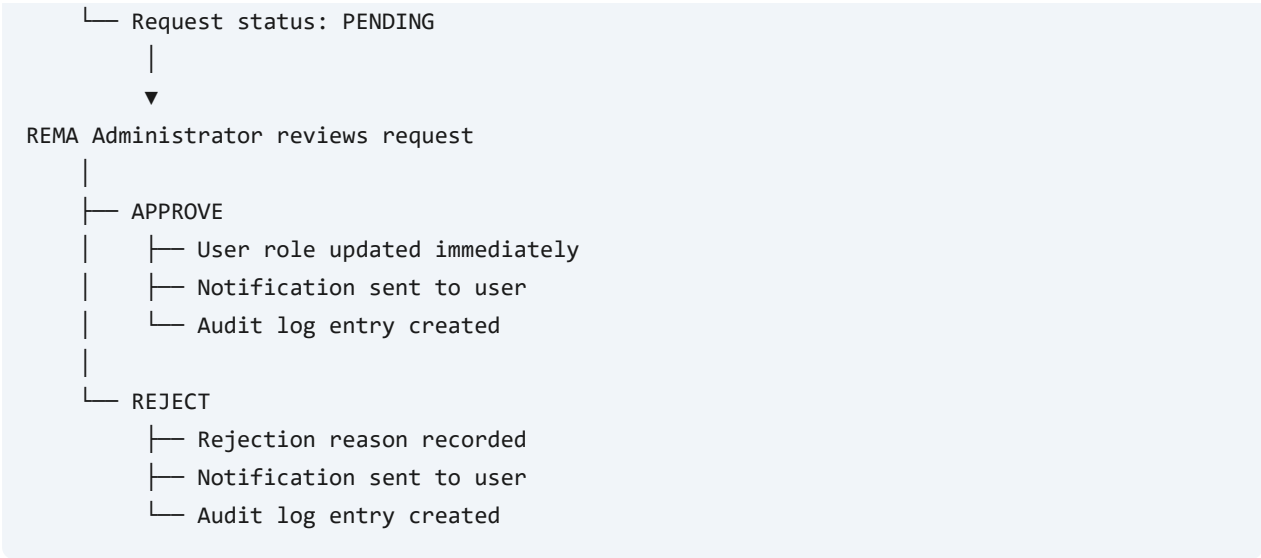
4.3 Suspension Controls

Action	Who Can Do It	Requirements
Suspend account	REMA Administrator	Must provide reason; optional end date
Set suspension end date	REMA Administrator	Date in the future
Auto-reactivate	System	Triggers when suspension end date passes
Deactivate account	REMA Administrator	Permanent; overrides suspension
Reactivate account	REMA Administrator	Manual action required

5. Role Change Workflow

5.1 Process Flow





5.2 Role Change Constraints

Constraint	Rule
Only REMA Administrators can approve role changes	Enforced by RBAC
Users cannot approve their own role change	System validation
All role changes are logged in the audit trail	Automatic trigger
Role changes take effect immediately upon approval	Database trigger
Previous role is recorded for audit purposes	Stored in request record

6. Data Access Controls (Row-Level Security)

6.1 Database-Level Enforcement

Data	REMA Admin	Ministry Reporter	District Officer	Policy Monitor	Dev Partner	Public Viewer
All user profiles	Read/Write	Own only	Own only	Own only	Own only	Own only
All reports	Read/Write/Delete	Read own + review all	Read own	No	No	No
Indicators	Read/Write	Read	Read	Read	Read	Read
Targets	Read/Write	Read	Read	Read	Read	Read
Audit log	Read	No	No	No	No	No
Notifications	Own	Own	Own	Own	Own	Own

Risk register	Read/Write	Read	No	Read	Read	No
Compliance records	Read/Write	Read	Read	Read	Read	No

6.2 Security Enforcement Stack

Layer 1: Client-Side Route Guards
ProtectedRoute component checks user.role
Prevents rendering unauthorised pages
Layer 2: Supabase Client (API Layer)
JWT token includes user identity
Every API call authenticated
Layer 3: PostgreSQL Row-Level Security
RLS policies evaluate auth.uid() and user role
Database refuses unauthorised data access
Cannot be bypassed via direct API calls

7. Default Role Assignment

Scenario	Default Role	Rationale
New user registration	policy_monitoring (Policy Monitor)	Safest default — read-only access
REMA designates administrator	dashboard_management	Manual assignment by existing admin
Ministry focal point onboarded	lead_government_ministry_reporting	Admin assigns after institutional verification
District officer onboarded	local_reporting	Admin assigns after district verification
Development partner registered	programme_alignment	Admin assigns after partner verification
Public access requested	public_viewer	Admin assigns or self-service

8. Audit & Accountability

8.1 Actions Logged

Action Category	Example Actions	Logged Fields
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Authentication	Login, logout, session refresh	User ID, timestamp, IP address
Data Submission	Report submitted (T01–T07)	User ID, tool type, district, timestamp
Verification	Report approved, report rejected	Reviewer ID, report ID, decision, reason
User Management	Role change, suspension, deactivation	Admin ID, target user, old/new state
Data Export	CSV/PDF/JSON export	User ID, export scope, timestamp
System Access	Page views for sensitive pages	User ID, page, timestamp

8.2 Audit Log Retention

Attribute	Value
Retention Period	Indefinite (append-only, never deleted)
Access	REMA Administrators only
Immutability	No UPDATE or DELETE operations permitted
Fields	User ID, action type, action description, detail, role at time of action, IP address, timestamp

9. Institutional Mapping

Institution	Recommended Role	Typical Users
REMA	REMA Administrator	Biodiversity department staff, IT team
Ministry of Environment	Lead Government Ministry Reporter / Policy Monitor	Focal points, senior advisors
Ministry of Agriculture	Lead Government Ministry Reporter	Sector reporting officers
Ministry of Lands	Lead Government Ministry Reporter	Land use reporting officers
Rwanda Development Board	Lead Government Ministry Reporter	Tourism & conservation officers
District Local Government	District Reporting Officer	Environmental officers (30 districts)
UNDP / GIZ / USAID	Development Partner	Programme managers, analysts
University of Rwanda	Development Partner	Research faculty, RBIS team
Private Sector / CSOs	Development Partner or Public Viewer	EIA compliance officers, NGO staff

General Public	Public Viewer	Citizens, journalists, researchers
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